

TEMPORAL INTEGRITY, ALIASING, AND RECONCILED PRESENTATION

Independent Deep-Research Pass and Simulation-Lab Development Framework

Central Research Question

At what point does a reconciled representation of time become materially different from the events from which it was produced; why does that divergence occur; can an observer detect and correct it; and what happens when the layer responsible for the error is no longer reachable?

Field	Value
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Safety posture	Benign engineered proxies; telemetry-first MVP; consent and formal review before human-participant research
Intended use	Simulation-lab planning, real-world bench testing, funding planning, scientific red-team review

Research boundary: Established signal-processing, perception, time-synchronization, camera, and modeling-and-simulation literature is distinguished from OAL internal terminology and speculative architecture. No OAL construct is presented as an established scientific finding.

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Recommended architectural refinement

Use Temporal Integrity or Temporal Presentation-State Integrity as the umbrella construct. Keep strict temporal aliasing as a mathematically defined mechanism within that larger family. Clock drift, buffering, postdiction, temporal binding, chronostasis, rolling shutter, and timestamp disagreement can all create timeline divergence without being aliasing.

1. Mission and Purpose

1.1 Mission

The mission of the Temporal Integrity research program is to determine when, how, and under what conditions a system composed of multiple observers, sensors, devices, clocks, buffers, renderers, networks, software processes, and human perceptual mechanisms produces a coherent presentation of time that materially differs from the best-supported sequence, frequency, duration, simultaneity, or causal ordering of the events that generated it.

The program then asks a second systems question: if a discrepancy is demonstrated, can the relevant layer identify the mechanism, reach the controlling layer, apply a correction, preserve the correction, and prevent recurrence? This makes temporal integrity a bridge among the existing OAL work on presentation-state divergence, interfaces and apertures, and Conditional Failure at the Root Level. [I1][I2][I4]

1.2 Purpose of this report

- Establish a scientifically defensible distinction between strict temporal aliasing and neighboring temporal phenomena.
- Preserve the user-originated central observation: the mind and every device it uses are each partial reconcilers, and the lived or displayed timeline is an output rather than a complete record.
- Map timing phenomena into CFTRL, the Department of Endings, Open Window v0.3, and PSIF without presenting OAL terminology as established science.
- Define an executable simulation laboratory with an authoritative event plane, independent audit plane, controlled faults, preregistered null hypotheses, and correction tests.
- Define real-world analogues that use triangulated evidence rather than claiming access to a universal physical-world ledger.
- Create a staged funding pathway, implementation gates, human-subjects boundary, critical blockers, red-team alternatives, and a concrete action plan.

1.3 Core scope statement

Scope rule

The research object is not "is perception false?" The research object is the measurable relationship among event production, acquisition, timestamping, transport, reconciliation, presentation, perception, action, and correction.

2. Executive Summary

The experienced present is not a raw recording of the world. It is an output assembled from signals with different latencies, integration windows, uncertainties, and update rules. Devices add additional temporal representations: cameras expose and sample; operating systems maintain multiple clocks; networks synchronize imperfectly; media pipelines buffer and decode; displays refresh; cloud services timestamp; and user interfaces decide which "now" to show. The person using those systems then integrates the device presentations with visual, auditory, proprioceptive, interoceptive, attentional, and memory processes.

Strict temporal aliasing is a well-defined signal-processing phenomenon. When the sampling rate is below the rate required to represent the relevant frequency content, higher-frequency structure can become indistinguishable from a lower-frequency reconstruction. The Nyquist rate is twice the highest frequency

in a band-limited signal, and sampling below that rate causes aliasing. Pre-filtering is required because once aliased components are present in the samples, the original identity generally cannot be reconstructed from those samples alone. [E1]

Temporal mismatch is broader than aliasing. Human timing judgments can be affected by postdiction, temporal binding, chronostasis, attentional limitations, and flicker fusion. Device timing can be affected by wall-clock jumps, drift, network asymmetry, buffering, rolling-shutter acquisition, and shared clock faults. These mechanisms must remain distinct if the laboratory is to diagnose rather than merely collect anomalies. [E4][E5][E8][E9][E10][E11][E12][E13]

The central systems insight is that a representation can be locally coherent, useful, and even smooth while remaining wrong about frequency, duration, sequence, simultaneity, or causal relationship. This is directly relevant to PSIF because a time label is itself a presentation claim. "This happened now," "A happened before B," "this lasted three seconds," and "this stream is live" are assertions that can be compared with an evidence plane. [I3][I4]

The strongest implementation strategy is therefore not a large chamber or human study. It is a staged program: build the temporal provenance schema and Python simulator; demonstrate a mathematically predicted alias under controlled conditions; prove clock independence and replay; add skew, buffering, rolling-shutter, and wall-vs-monotonic controls; then test inaccessible-root timing failure. Only after those foundations survive should the program add high-fidelity instrumentation or formal human-participant work.

3. Research Basis and Evidentiary Boundary

3.1 Three evidence classes

Class	Meaning	Examples in this report
Established / standard technical basis	Well-defined theory, published empirical result, or authoritative technical standard/documentation.	Nyquist sampling; NTP asymmetry limits; PTP synchronization; Android wall vs monotonic clocks; rolling shutter; NASA M&S credibility practices.
Active / contested research	Published or emerging scientific mechanisms where interpretation remains debated or evidence is incomplete.	Continuous wagon-wheel models; attention-dependent discrete sampling; neural delay model; breadth and mechanism of postdiction across paradigms.
OAL internal architecture	User-developed design language and simulation constructs. These are treated as conceptual/test architecture, not as scientific fact.	CFTRL; Department of Endings; Open Window Doctrine; CETE codes; ONESTAMP; PSIF-specific labels; Temporal Endgame Desk.

3.2 Why the distinction matters

The most important scientific correction to the existing temporal draft is linguistic rather than conceptual: a human brain should not be described literally as a conventional clocked analog-to-digital converter with a fixed Nyquist frequency. That analogy is useful for systems intuition, but the empirical literature supports multiple temporal mechanisms with substantial dependence on task, modality, attention, stimulus conditions, and individual differences. Continuous wagon-wheel research has supported attention-linked periodic sampling models, but alternative accounts remain live, and a 2026 delay-coupled neural-field paper is explicitly a preprint. [E6][E7]

The laboratory should therefore measure participant behavior instead of assigning a universal human frame rate. This preserves the original observation - that awareness is an incomplete reconstruction - while avoiding a stronger neural mechanism claim than the literature supports.

4. The Central Phenomenon: Reconciled Presentation

4.1 A top-down view

From a top-down perspective, a modern environment contains many partial views trying to reconcile into one operational account. A person may simultaneously rely on a phone, smartwatch, television, computer, cameras, speakers, notifications, cloud services, and physical sensory cues. Each device has its own clocks, acquisition policy, buffering, refresh cadence, synchronization source, and interface state. The human observer has no direct access to a complete pre-perceptual record of the surrounding world; the observer acts from the representation made available to awareness.

This creates a stack of reconcilers rather than one camera-versus-reality problem. The key risk is not simply that one component is delayed. The larger risk is that multiple individually reasonable reconstructions can compose into a timeline that is internally consistent but globally wrong about order or causation. The existing OAL temporal draft explicitly identifies this composition problem and connects it to technology dependence. [I1]

4.2 Recommended truth vocabulary

Term	Recommended meaning
First-person operational truth	The representation available to the participant and therefore the state from which the participant actually acts.
Evidenced state	The best-supported state reconstructed from independent observations, records, mechanisms, and provenance.
Simulation ground state	The authoritative state-transition record intentionally exposed by a simulator for a defined experiment.
Audit reference	A timing or evidence channel engineered outside the fault domain under test.
Causal history	Event dependencies or happens-before relationships, not merely wall-clock order.

Important distinction

A percept can be operationally authoritative to the person without being treated as universal physical ground truth. Conversely, a simulation ledger can be authoritative for one experiment without being assumed infallible during a root-ledger failure experiment.

5. Scientific Research Account

5.1 Strict temporal aliasing

MIT signal-processing materials summarize the sampling theorem directly: a band-limited signal can be sampled without loss only when the sampling frequency exceeds twice the highest relevant frequency;

sampling below the Nyquist rate produces aliasing. Components above the folding frequency can masquerade as lower-frequency content after sampling. [E1]

For a sinusoidal component with frequency f sampled at f_s , aliases are separated by integer multiples of f_s . A useful laboratory expression is: $f_{\text{alias}} = |f - k \cdot f_s|$, with k selected so the apparent frequency lies in the sampled baseband. For periodic spatial patterns such as rotating spokes, the same principle can produce apparent reverse motion when successive frames support a shorter displacement interpretation in the opposite direction.

The engineering implication is decisive: a true alias experiment should predict the alias before the run. A striking visual reversal alone is not sufficient. The run should demonstrate that the measured apparent frequency or direction follows the predicted fold, that increasing sample rate moves or eliminates the fold, and that an appropriate pre-sampling filter suppresses it.

5.2 Spatio-temporal aliasing in simulator displays

NASA has explicitly studied spatio-temporal aliasing in out-the-window flight-simulator visual systems. NASA describes the artifact as a motion-quality problem that can affect motion perception and simulator task performance, not merely as a visual curiosity. This gives the OAL program a strong real-world anchor: temporal presentation errors can matter when a user acts on a rendered world. [E2]

5.3 Continuous wagon-wheel research: useful, but contested

Under continuous illumination, some observers report apparent motion reversals. VanRullen and colleagues reported strong frequency dependence around 10 Hz and modeled the effect using attention-dependent temporal subsampling in roughly the 10-20 Hz range. Later work notes that alternative motion-detector/adaptation explanations remain possible. The correct conclusion for this program is therefore not "the brain refreshes at 13 Hz," but "some motion-perception phenomena show temporal structure that can be experimentally parameterized and may involve attention-dependent sampling or other dynamics." [E6]

A 2026 Parks and Kilpatrick preprint proposes a delay-coupled neural-field mechanism capable of quantized propagation speeds and reversals. Because it is a preprint, it should be treated as an emerging mechanistic alternative, not a settled replacement for earlier models. [E7]

5.4 Postdiction and flash-lag

Eagleman and Sejnowski reported that the flash-lag percept can depend on events in approximately the 80 ms after the flash, motivating a postdictive rather than purely predictive or online account. Later reviews discuss postdiction as a broader family of retrospective perceptual effects. This supports one element of the user-originated intuition: the percept attributed to an earlier moment can depend on information that arrives later. It does not justify a single universal 80-500 ms brain buffer. [E8]

5.5 Multisensory temporal binding

Review literature on multisensory temporal binding shows that observers can report audiovisual events as simultaneous even when onset times differ by 100 ms or more under some conditions. The binding window varies across people, tasks, stimulus properties, and modality relationships. For the laboratory, this becomes an empirical participant parameter rather than a fixed architecture constant. [E9]

5.6 Chronostasis and saccadic temporal reconstruction

Chronostasis is the stopped-clock illusion in which the first visual interval after a saccade appears subjectively prolonged. Yarrow and colleagues described this as a backward extension of the post-saccadic

percept that helps preserve continuity across the visual disruption caused by an eye movement. Later work shows temporal distortions around saccades can involve multiple effects. Chronostasis is therefore a duration-reconstruction phenomenon, not strict aliasing. [E10]

5.7 Attentional blink

In rapid serial visual presentation, the ability to report a second target is often reduced when it appears approximately 200-500 ms after the first target. This is a useful neighboring phenomenon because an event may be physically present and accurately rendered yet fail to become part of the participant-accessible event history under specific attentional conditions. It should be classified as attentional temporal loss rather than aliasing. [E11]

5.8 Critical flicker fusion

Critical flicker fusion is not one universal number. Reviews report broad ranges and identify strong dependence on stimulus intensity, retinal location, wavelength, ambient lighting, viewing geometry, individual condition, and measurement method. This is another reason not to convert a perceptual threshold into a single human "frames per second" specification. [E12]

5.9 Network and operating-system time

NIST identifies network-path asymmetry as a dominant error source in wide-area NTP time transfer, typically limiting uncertainty to about one millisecond in the referenced study. IEEE 1588 Precision Time Protocol is designed for much tighter synchronization and supports sub-microsecond performance in engineered networks. The existence of better synchronization controls does not eliminate temporal-integrity questions; it provides experimental controls and a path to quantify uncertainty. [E3][E4]

Operating systems already distinguish clock semantics. Android documents that wall time can jump backward or forward and recommends monotonic clocks such as `elapsedRealtime` for interval measurement. Linux similarly distinguishes `CLOCK_REALTIME`, which can jump with time adjustments, from `CLOCK_MONOTONIC`, which does not experience discontinuous wall-clock jumps. These are direct real-world examples of why a timestamp is not self-describing evidence. [E5][E21]

5.10 Rolling shutter and event-based sensing

Rolling-shutter sensors expose/read rows at different times; Hamamatsu explicitly notes that exposure timing differs across the same screen. A nominally single image can therefore encode multiple moments. Global-shutter capture provides a useful comparison because all rows share the same exposure timing. [E13]

Event cameras provide a further contrast: rather than fixed-rate frames, they asynchronously report per-pixel brightness changes with timestamps and can achieve microsecond-scale temporal resolution. They are not a universal solution, but they are a powerful control architecture for determining how much temporal distortion is introduced by fixed-frame acquisition. [E14]

6. Recommended Temporal Taxonomy

The taxonomy below should replace any practice of labeling all timing discrepancies as temporal aliasing. Codes are proposed for laboratory use; they do not claim to be industry standards.

Code	Phenomenon	Mechanism / observable	Strict alias?
TA	Temporal aliasing	High-frequency structure reconstructs as a lower-frequency or reversed identity because sampling is insufficient.	Yes
SK	Clock skew / drift	Clocks disagree in offset or rate for the same event.	No
JI	Jitter	Sampling or transport intervals vary.	No
BF	Buffering / batching	Events are held and released later as a group.	No
DR	Drop / repetition	Samples are lost, repeated, or held.	No
RS	Rolling-shutter composite	Different parts of one nominal frame represent different moments.	No
OI	Order inversion	Presented order conflicts with evidenced happens-before relation.	Not necessarily
DD	Duration distortion	Presented interval differs materially from evidenced interval.	No
TB	Temporal binding	Asynchronous sensory events are treated as simultaneous/integrated.	No
PD	Postdiction	Later information alters percept attributed to an earlier moment.	No
CH	Chronostasis	Post-saccadic duration is overestimated.	No
AB	Attentional temporal loss	A temporally proximate event has reduced access/reportability.	No
CFF	Flicker fusion	Temporal modulation is perceived as steady beyond task-dependent threshold.	No
MX	Mixed / interacting	Two or more temporal mechanisms jointly produce divergence.	Possibly

6.1 Umbrella term recommendation

Recommended umbrella

Temporal Presentation-State Divergence (TPSD): a material inconsistency between the presented timeline and the best-supported event timing, ordering, frequency, duration, simultaneity, or causal history. TPSD is neutral about mechanism and intent. Temporal aliasing is one subtype.

7. Unified Top-Down Model

The full stack can be represented as a closed loop rather than a one-way sensor chain:

Core loop

EVENT -> ACQUISITION -> LOCAL TIME -> TRANSPORT -> RECONCILIATION -> PRESENTATION -> HUMAN
 RECONCILIATION -> ACTION -> NEW EVENT

7.1 What each stage can change

Stage	Temporal transformation	Example failure
Event production	Native event rate and causal dependencies.	Event rate exceeds downstream sampling budget.
Acquisition	Exposure/integration window and sample schedule.	Aliasing, rolling shutter, dropped samples.
Local time	Timestamp semantics and oscillator behavior.	Skew, drift, wall-clock jump.
Transport	Latency, reordering, jitter.	Late arrival presented as current.
Reconciliation	Merge streams and choose a coherent order.	Last-write-wins overrides causal order.
Presentation	Refresh, buffering, labels, "live" indicators.	Smooth but stale display.
Human reconciliation	Attention, binding, postdiction, expectation.	Separated events experienced as simultaneous.
Action	Decision based on presented state.	Incorrect control action based on wrong order.
Correction	System recognizes and repairs mismatch.	Presentation persists despite corrected state.

7.2 Coherence is not correctness

A central metric should explicitly separate presentation coherence from temporal fidelity. In ordinary product design, smoothing, interpolation, buffering, animation, and latency hiding are often desirable because they reduce visible discontinuity. Yet the same methods can preserve the feeling of continuity while increasing the gap between what is shown and when the underlying events occurred. This is the temporal form of the PSIF distinction between event fidelity and causal fidelity. [I3]

8. Incorporation into Existing OAL Frameworks**8.1 Conditional Failure at the Root Level**

The CFTRL source asks what happens when a nested or stacked system scales until the root fails while lower tiers cannot access or repair the root. It specifically calls for computational modeling, stress-to-failure, redundancy from initialization, and logging the scale at which the architecture breaks. [I2]

Temporal integrity supplies a concrete CFTRL failure family: root clock failure, root sampling-policy failure, reference-source loss, causal-ledger failure, or timing-maintenance abandonment can allow lower layers to continue producing a coherent presentation while temporal truthfulness degrades. This is more informative than a simple crash because the system may remain usable enough to hide the failure.

8.2 Department of Endings

The source defines the Department of Endings as the operational arm that runs failures, logs what failed and under what conditions, extracts lessons, and feeds them back. A Temporal Endgame Desk is therefore a reasonable laboratory implementation: systematically run reverse-motion folds, order inversions, duration distortions, root-clock loss, last-write-wins corruption, buffer runaway, and correction failures. [I2]

8.3 Open Window Doctrine v0.3

Open Window v0.3 defines window/aperture/interface broadly as any vector across which observation, state, information, influence, access, timing, control, or interaction may cross; it makes the Interface Airlock optional; uses CETE as the parent transfer term; retains MU/MX for unexpected classes; and limits MVP tests to declared interfaces and benign engineered proxies. [I5][I6]

Recommended temporal rule: not every temporal discrepancy is a CETE. A camera that aliases its own wheel is a local acquisition phenomenon. A CETE-temporal label should be reserved for a timing/order/duration effect specifically studied across a declared interface or vector. This prevents the Open Window vocabulary from swallowing ordinary signal-processing failures.

8.4 Presentation-State Integrity Framework

PSIF treats presentation as a claim about state, separates state, event history, observation, presentation, interpretation, and reconciliation, and requires evidence to be graded without collapsing occurrence, attribution, causation, coordination, or intent. It also explicitly warns against over-correlation, perspective overreach, sensor monoculture, and treating a complete-looking log as infallible truth. [I3][I4]

Temporal Integrity extends that architecture by recognizing timing itself as a claim. "Now," "first," "simultaneous," "live," and "lasted three seconds" are all presentation assertions that can be tested against event provenance and uncertainty.

8.5 Incorporation map

Framework	Temporal role	Primary lab question
CFTRL	Inaccessible timing/root failure.	What happens when the timing authority above a layer fails and cannot be repaired from below?
Department of Endings	Failure execution and learning.	Which temporal endgames fail, at what scale, and what design change prevents recurrence?
Open Window v0.3	Interface/vector and CETE classification.	Does timing/order/duration change across a declared interface, and is the effect local or cross-boundary?
PSIF	Presentation vs evidenced state and causal fidelity.	Does the presented timeline correspond to independently preserved event history?
Temporal Integrity	Mechanism-specific time architecture.	Which timing mechanism produced the divergence, how material is it, and can it be corrected?

9. Simulation-Lab Architecture

9.1 Recommended L0-L5 timing stack

Layer	Name	Timing object / responsibility
L0	Independent audit plane	Reference timing outside the intended fault domain; experiment control; immutable evidence; hashes.
L1	Engine / canonical state	Simulation tick; authoritative event IDs; state transitions; parent/dependency IDs.
L2	Environment processes	Physical/simulated dynamics at native event rates.
L3	Device / aperture objects	Per-device sample rate, exposure, clock source, buffering, refresh, network behavior, power state.
L4	Occupant reconciler	Participant report channel; binding/postdiction/attention parameters only when human research is authorized.
L5	Presentation / HUD	Displayed order, duration, status, "now," "live," and narrative claims.

For timing work, this stack is preferable to arguing over historic Open Window layer counts because it assigns every temporal transformation to a testable function. Other Open Window diagrams can remain in force for non-temporal work until a later global merge. [I1]

9.2 Verification and validation posture

The lab should follow the spirit of NASA-STD-7009B without implying NASA endorsement. NASA-STD-7009B requires modeling-and-simulation acceptance criteria to be defined for intended use and approved by appropriate technical authority; the 2026 NASA handbook provides implementation guidance for good M&S practices and credibility assessment. [E15][E16]

- Verification question: Did the simulator, clock models, buffers, and fault injectors implement the specified design correctly?
- Validation question: Does the model behave sufficiently like the selected referent for the intended decision?
- Credibility question: Are limitations, uncertainty, sensitivity, evidence provenance, and intended use explicit enough to support the decision being made?

10. Temporal Provenance and Independent-Oracle Requirements

10.1 Temporal provenance overlay

Every material event should preserve raw and derived timing separately. Time normalization must never destroy the evidence required to audit the normalization itself.

Field group	Minimum fields
Identity	event_id; parent_event_id; condition_id; source_id; process_id
Raw timing	source_time; capture_start; capture_end; local_monotonic_time; local_wall_time

Field group	Minimum fields
Pipeline timing	ingest_time; decode_time; queue_release_time; presentation_time; participant_report_time
Clock metadata	clock_id; clock_type; reference source; estimated uncertainty; last sync; stratum/profile where applicable
Ordering	sequence_number; dependency IDs; happens-before links
Integrity	raw hash; trace hash; software version; configuration hash; random seed where applicable
Derivation	normalized_time; normalization method; transform version; derived flag
Evidence grade	recorded / derived / correlated / inferred / predicted / claimed / unresolved

10.2 Preserve raw, derive corrected

Time-normalization rule

A time airlock may flag, buffer, translate, or refuse timestamps that violate a declared skew budget, but it must not overwrite the original timing record. Store raw time first; store corrected/normalized time as a derived field with explicit provenance.

10.3 Independent Oracle Requirement

If L0 and L1 are produced by the same clock, oscillator, time service, logging thread, random seed, or failure domain, agreement between them may be a sensor-agreement illusion. For ordinary presentation experiments the engine ledger can serve as canonical state. For CFTRL root-failure experiments, at least one audit reference must be outside the fault domain being stressed.

Hard rule

If the experiment cannot engineer an independent reference outside the injected failure domain, it may still test local presentation behavior, but it cannot support a root-failure claim.

11. Mathematical and Computational Model

11.1 Nyquist margin

For a highest relevant band-limited frequency B and sample rate f_s , define a laboratory Nyquist Margin: $NM = f_s / (2*B)$. $NM > 1$ means the theoretical sampling boundary is exceeded; $NM < 1$ means relevant content is necessarily undersampled. Practical systems require margin beyond the ideal boundary because filters have transition bands and real signals are not perfectly band limited. [E1]

11.2 Alias prediction

For a sinusoidal component, predict the apparent baseband frequency with $f_{\text{alias}} = |f - k*f_s|$ for an integer k selected to map into the sampled baseband. For rotational periodic patterns, record angle per

engine tick and compute expected nearest-equivalent spoke displacement under each sampling condition. The prediction must be registered before the observer or algorithm output is scored.

11.3 Minimum simulator objects

Object	Minimum parameters
RootClock	frequency; drift_ppm; offset; step_fault; maintenance_alive; reference_id
LayerSampler	fs; jitter distribution; exposure/integration; buffer depth; drop policy; anti_alias_filter
EventProcess	true frequency / rate; causal parents; optional periodic spatial structure
Transport	latency distribution; asymmetry; reordering; batching policy
Reconciler	ordering rule; lowest-frequency/smoothness prior; binding window; last-write-wins flag
Presentation	refresh rate; interpolation; frame queue; stale-state policy; "live" label
Ledger	append-only event ID; root/local times; dependencies; hashes; version metadata

11.4 Failure injections

- Disable root clock maintenance while lower layers continue sampling.
- Increase event frequency while holding sample rate fixed until a predicted fold occurs.
- Add pairwise drift until wall-clock order conflicts with happens-before order.
- Inject a wall-clock step while preserving monotonic time.
- Increase buffer depth until smooth presentation remains stable but source-to-presentation latency crosses materiality threshold.
- Create shared-clock failure so nominally independent sensors agree on the same wrong time.
- Toggle anti-alias prefilter at design time to quantify prevention versus post-hoc detection.
- Corrupt the canonical event ledger in a CFTRL-specific campaign while maintaining an external audit oracle.

11.5 Primary CFTRL result

Graceful Presentation / Degraded Truth

The most important CFTRL temporal failure may be the condition in which presentation coherence remains high while temporal fidelity drops below tolerance. The system does not visibly crash; it continues operating while losing reliable knowledge of when events occurred.

12. Executable Simulation Scenario Program

All scenarios use the Open Window implementation memo schema: purpose, declared interface objects, proxy class, airlock condition, exact setup, null hypothesis, positive and negative controls, metrics, stop conditions, replications, replay, recurrence, preserved data products, and post-run classification. [I6]

12.1 TA-00 - Temporal Identity and Calibration

- Purpose: prove the measurement system can establish known simultaneity, known offset, known frequency, known causal order, and timestamp uncertainty before anomaly testing begins.
- Setup: hardware or simulated trigger distributes a known pulse to all channels; audit oracle records trigger time and expected arrival topology.
- Null: every declared channel remains within its registered uncertainty and skew budget.
- Positive control: deliberately offset one channel by a known value; the system must detect and quantify the offset.
- Negative control: synchronized channels with no injected fault.
- Pass condition: calibration error and measured uncertainty are below scenario-specific materiality thresholds; otherwise all downstream cards are blocked.

12.2 TA-01 - Undersampled Motion (strict alias)

- Purpose: demonstrate a quantitatively predicted frequency/direction alias.
- Setup: periodic rotating target; true angle stored each engine tick; camera/display sample rate swept across declared spoke-interval ratios.
- Null: presented direction/frequency matches engine history within tolerance.
- Positive control: a condition below the required sampling rate with preregistered predicted fold.
- Negative control: adequate sample rate and/or pre-sampling anti-alias control.
- Primary metrics: TA_fold, FAE, NM, D_ps_t, replay concordance.
- Required result: observed alias must match the preregistered mathematical prediction and change/disappear under the declared intervention.

12.3 TA-02 - Anti-Alias Intervention

Repeat TA-01 with the event process held constant while only sample rate and pre-sampling filtering are manipulated. This card distinguishes a mechanism-confirming intervention from a visually impressive demonstration. If the supposed alias persists unchanged when the predicted control should remove it, the mechanism is not established.

12.4 TI-03 - Multi-Device Now

- Purpose: test skew, binding, and ordering across two or more device presentations.
- Setup: inject 0, 25, 50, 100, 200, 500, and 1000 ms offsets across channels, randomized across trials.
- Conditions: no synchronization; software/network time discipline; hardware-stamped reference where available.
- Metrics: SK_max, OI_rate, source-to-presentation latency, participant binding width if human reports are authorized.
- Interpretation: disagreement is clock/binding divergence unless a true sampling fold is separately demonstrated.

12.5 TI-04 - Buffer / Batch / Stitch

- Purpose: test whether buffering can preserve apparent continuity while changing event timing and grouping.
- Setup: event stream with known sequence and intervals; progressively increase queue depth and burst release.
- Null: presentation preserves declared interval/order tolerance.
- Metrics: BF latency, P99 jitter, CCS coherence-correctness separation, order-inversion rate.

- Correction: display provenance or age-of-data indicator; causal ordering rather than last-write-wins.

12.6 TI-05 - Rolling-Shutter Temporal Composite

Render a moving timing grid to simulated rolling- and global-shutter sensors. The same nominal frame should carry line-specific timestamps for rolling shutter. Score whether a downstream algorithm incorrectly treats the frame as a single instant. Add an event-camera analogue in a later campaign to compare asynchronous sensing.

12.7 TI-06 - Wall Time vs Monotonic Time

Inject a wall-clock jump inside an isolated software environment while preserving monotonic elapsed time. Determine whether event duration, retry logic, sequence ordering, timers, and logs follow the correct clock semantics. Android and Linux documentation make this a highly practical real-world analogue. [E5] [E21]

12.8 OW-TI-01 - Black-Screen Residual Timing State

Translate the original Open Window unattended-aperture idea into a timing/state experiment: compare screen-on, display-dark-but-powered, sleep, offline/airplane, and fully powered-down states. The finding of residual timers, packets, wakeups, or heartbeats establishes a presentation-state versus operational-state difference. It does not by itself establish strict aliasing, cross-environment transfer, or metaphysical infection.

12.9 CFTRL-TI-01 - Root Maintenance Loss

- Begin with all layers synchronized and calibrated.
- Disable or degrade root timing maintenance while preserving lower-layer operation.
- Prevent child layers from repairing or directly querying the root timing policy.
- Increase scale: number of devices, sample rates, buffers, and independently drifting clocks.
- Measure the first point at which temporal fidelity crosses a materiality threshold while presentation coherence remains acceptable.
- Apply design-time redundancy in a second condition: independent audit clock, monotonic traces, causal IDs, and anti-alias filtering.
- Compare whether redundancy prevents failure, delays failure, or merely improves detection.

13. Real-World Test Program

13.1 Translation rule

The physical world normally does not provide a universal engine ledger. PSIF already solves this problem by replacing universal ground truth with best-supported evidenced state reconstructed from independent records, repeatable procedures, timestamps, instruments, observations, and interventions. The same rule should govern Temporal Integrity. [I3]

Minimum real-world evidence rule

Use at least two meaningfully independent timing channels for material claims, document common-mode dependencies, preserve raw traces, pre-register tolerance, and allow "unresolved" when uncertainty intervals overlap.

13.2 RW-01 - Rotational alias benchmark

Use an enclosed motorized periodic target with encoder-reported angular position. Record it simultaneously with a conventional frame camera at multiple frame rates and a higher-speed/global-shutter reference. If available, add an event camera. Predict the alias in advance and compare observed direction/frequency against encoder history.

13.3 RW-02 - Multi-device simultaneous trigger

A hardware trigger produces a common LED and/or audio pulse captured by phone, computer, television/capture chain, and independent reference channel. Record device wall time, monotonic time where accessible, capture time, presentation onset, and reference time. The goal is to quantify separate "nows" in one room, not to assume that disagreement implies deception.

13.4 RW-03 - Clock-step software sandbox

Use a controlled test application or virtual machine. Adjust or simulate wall-clock time while preserving monotonic elapsed time. Verify whether durations, retries, timers, event order, and logs remain valid. This experiment can be run without modifying production devices or systems.

13.5 RW-04 - Rolling vs global shutter

Use a high-speed moving marker or coded timing pattern. Compare rolling and global shutter. Score spatial distortion, inferred motion, and whether downstream processing incorrectly assumes one timestamp for the entire rolling-shutter frame. Hamamatsu documentation provides a direct technical reference for line-dependent exposure timing. [E13]

13.6 RW-05 - Device power/presentation states

On a lab-owned device and isolated SSID, capture packet, timer, and wake activity under declared states: active display, display dark, sleep, offline, and powered down. Preserve privacy by excluding third-party content and accounts. The outcome is an operational-state map, not a claim that a dark screen is universally "open."

13.7 RW-06 - Incident reconstruction drill

Create a benign synthetic incident with a known event sequence and distribute evidence across camera, application log, network trace, badge-like event, and operator note. Introduce controlled timestamp skew. Ask independent reviewers to reconstruct order first from wall time and then from causal/provenance links. This directly tests whether timestamp agreement or provenance is the more reliable route to causal history.

14. Human-Perception Research Program

14.1 Why human research is a separate campaign

Human perception is central to the user-originated phenomenon, but it is also the easiest place to overclaim mechanism and the most administratively complex place to begin. The recommended MVP remains telemetry-first. Human work should start only after the device and simulation timing harness is stable and an appropriate human-research determination has been obtained.

14.2 HP-01 - Audiovisual simultaneity judgment

Present audio and visual events at randomized onset offsets around objective simultaneity. Estimate each participant's psychometric probability of reporting simultaneity and derive a temporal binding window rather than assigning a universal threshold. The multisensory literature supports this as a standard way to characterize temporal tolerance. [E9]

14.3 HP-02 - Chronostasis

Use a standard stopped-clock-style saccade paradigm to quantify subjective duration expansion after gaze shifts. This tests duration reconstruction and continuity, not aliasing. [E10]

14.4 HP-03 - Continuous motion reversal

If this campaign is pursued, treat continuous wagon-wheel reversals as an empirical perceptual response with competing mechanistic interpretations. Do not describe a positive result as proof of a global perceptual frame rate. [E6][E7]

14.5 Human-subjects governance

For U.S. research governed by the Common Rule, HHS regulations at 45 CFR part 46 provide baseline requirements for IRBs, informed consent, and compliance. HHS describes legally effective informed consent as a central protection when required, subject to exemptions or documented waivers. The project should obtain an appropriate formal determination rather than inventing an "IRB-equivalent" process. [E17]

- Prospective informed consent where required; clear description of stimuli and recording.
- Right to withdraw without penalty.
- No covert recording or undeclared stimulation.
- No clinical or diagnostic claims based on laboratory timing responses.
- Participant-level statistics; repeated trials within one person are not independent participants.
- Pre-registered stop criteria for discomfort, simulator sickness, or adverse effects.

15. Metrics and Statistical Design

15.1 Core metric dictionary

ID	Metric	Definition / use
D_ps_t	Temporal presentation-state discrepancy	Material timeline mismatches divided by audited events.
TA_fold	Alias-fold count	Reconstructions matching the predicted alias rather than source.
NM	Nyquist margin	$f_s / (2*B)$ for defined band-limited test content.
FAE	Frequency alias error	Difference between predicted and observed alias frequency/direction.
OI_rate	Order inversion rate	Presented pairs whose order conflicts with established happens-before relation.
SK_max	Peak pairwise skew	Maximum absolute time difference across declared clocks.
SPL	Source-to-presentation latency	Elapsed time from evidenced source event to participant-facing presentation.

ID	Metric	Definition / use
P99-j	99th percentile jitter	Tail variation in sample/transport/presentation timing.
TPU	Temporal provenance uncertainty	Uncertainty associated with a timestamp after reference, path, and instrumentation limits.
TBW_occ	Observed binding width	Largest tested asynchrony still commonly judged simultaneous for a participant/task.
TTD_t	Time to detect	Latency from evidenced mismatch to operator detection.
TTC_t	Time to correct	Latency from detection to validated repair or documented corrective-access failure.
CCS	Coherence-correctness separation	Degree to which presentation remains smooth/coherent while temporal fidelity degrades.
CMR	Common-mode risk	Share of nominally independent channels with a shared clock/failure source.
REP	Replay concordance	Whether independent replay reproduces the discrepancy under declared conditions.
COR-P	Correction persistence	Whether the repaired timing state remains corrected across follow-up trials.

15.2 Uncertainty-aware ordering

Store each timestamp conceptually as (time, uncertainty, clock_id). If uncertainty intervals overlap and the causal graph does not resolve order, the proper result may be "order unresolved." This is preferable to inventing a precise sequence from wall-clock values that cannot support it.

15.3 Statistical regimes

Deterministic engineering tests

Strict aliasing is primarily a prediction-and-measurement problem. The key question is whether the observed fold matches the mathematical prediction and responds correctly to sample-rate or filter intervention. Replications characterize repeatability, instrumentation error, and implementation defects; statistical significance is not the center of the claim.

Stochastic system tests

Jitter, network timing, scheduling, queueing, and reordering require randomized trial order, distributions, confidence intervals, tail metrics, and predeclared thresholds. Materiality should be defined by the use case, not by statistical detectability alone.

Human psychophysics

Human work should use participant-level models, psychometric curves, confidence intervals, and an a-priori sample-size plan informed by pilot variability. Hundreds of trials from one participant characterize that participant; they do not substitute for a human sample.

16. Gates, Acceptance Criteria, Safety, and Governance

16.1 Gate structure

Gate	Question	Failure disposition
G0	Is the schema complete, null written, proxy/interface declared, materiality threshold defined?	No run.
G1	Is the timing reference independent enough for the intended claim, with uncertainty characterized?	Block or downgrade claim.
G2	Do positive and negative controls behave as specified?	Revise instrumentation/card.
G3	Does the primary metric move in the preregistered direction under the manipulation?	Inconclusive or falsified; do not relabel.
G4	Does independent replay of preserved traces reproduce the effect?	Candidate only.
G5	Does a fresh acquisition independently recur?	Hold promotion.
G6	Are after-action, exceptions, limitations, and review recommendation complete?	Not eligible for ratification.
G7 - proposed	Does the identified correction reduce the discrepancy and persist?	Mechanism detected but correction incomplete.

16.2 Candidate Interface Event discipline

The Open Window implementation memo already provides the correct exception rule: an unexplained undeclared vector receives a unique ID, raw telemetry is preserved, the event is not promoted in the same run, and post-run disposition distinguishes instrumentation anomaly, unresolved candidate, replay-required candidate, or later-campaign candidate. Promotion requires independent observation, replay, recurrence, and intervention where possible. [I6]

16.3 Safety posture

- Benign engineered proxies only in MVP.
- No biological-harm framing as an experimental objective.
- No unaware-operator condition in MVP.
- Lab-owned devices and isolated networks for packet/timing capture.
- Stop conditions use numeric thresholds rather than adjectives.
- Raw evidence is preserved before any normalization, cleanup, or visualization.
- Null hypotheses are allowed to win; a clean falsification is a successful research outcome.

16.4 Authority and ONESTAMP

The attached implementation memo recommends a defined review group with the originator retaining final ratification authority. It also recommends that ONESTAMP eligibility occur only after scenario-card approval, gate disposition, after-action review, exception disposition, and written review recommendation. This governance structure should be locked before live execution. [I6]

17. Funding Pathway

The existing temporal draft contains useful planning bands. They should be treated as rough 2026 planning estimates rather than bids, and each tier should purchase a specific increase in evidentiary capability rather than simply more hardware. [I1]

Tier	Planning range	Primary scope	Evidence unlocked
T0 - Architecture/model	\$6,000-\$18,000	Taxonomy; provenance schema; Python stack; TA-00/TA-01 cards; review package.	Decision whether bench execution is justified.
T1 - Lean bench MVP	\$12,000-\$35,000	Strict alias benchmark; software skew/buffer tests; one lab-owned device; replay archive.	G0-G4 evidence or clean falsification.
T2 - Independent instrumentation	\$40,000-\$90,000	Independent hardware timing reference; two acquisition paths; isolated network; recurrence.	G5 recurrence; lower common-mode clock risk.
T3 - High-fidelity campaign	\$150,000-\$280,000	PTP-aware timing, multi-device capture, larger archive, optional participant campaign after approval.	Broader external validation only after T1/T2 survive.

17.1 Funding principle

Evidence unlocks fidelity

Do not fund T3 infrastructure to compensate for an unwritten null hypothesis, shared clock architecture, or missing provenance. Higher fidelity is justified only when lower-cost experiments survive preregistered falsification attempts.

17.2 Commercial / instrumentation pathway

The strongest commercialization narrative is not an abstract doctrine. It is a Temporal Integrity Test Harness or Temporal Provenance Instrumentation Platform that can inject known timing faults, preserve independent timing evidence, score presentation-state divergence, and test correction across simulations and multi-device systems.

NSF currently lists SBIR/STTR solicitations including a 2026 pilot emphasis on scientific instrumentation. Phase I applicants must first submit a Project Pitch and receive an invitation before a full proposal. NSF lists November 4, 2026 as the next full-proposal deadline for invited applicants under the current schedule. [E18][E19]

17.3 Academic / HCI pathway

For the human-device portion, an academic collaboration may fit human-centered computing. NSF describes HCC as research integrating computing with social and behavioral sciences to understand how human, technical, and contextual aspects of systems interact and to assess computing-system effects and risks. A proposal should focus on a narrow research question such as multi-device temporal reconciliation

or provenance-aware interface design rather than attempting to fund the entire OAL architecture at once. [E20]

17.4 Recommended funding sequence

1. Complete T0 with internal/seed/in-kind resources.
2. Produce one reproducible strict-alias benchmark and one real-world timing-divergence demonstration.
3. Document commercial use cases: incident reconstruction, simulation QA, robotics, displays/cameras, distributed operations, and safety interfaces.
4. Submit an NSF Project Pitch only after the technical innovation and technical risk are crisp enough to distinguish the tool from existing observability/synchronization products.
5. Use T2/T3 only after the project has a validated need for independent hardware timing or human research.

18. Critical Implementation Blockers

Blocker	Required resolution
Scientific terminology	Remove literal claims that the brain has a universal frame rate or conventional Nyquist limit.
Umbrella taxonomy	Adopt Temporal Integrity/TPSD above strict aliasing.
Independent reference	Define how L0 remains outside each injected fault domain.
Provenance schema	Implement immutable raw timing plus derived normalized timing.
Calibration	TA-00 must precede anomaly scenarios.
Materiality	Define task-specific thresholds: a measurable 1 ms difference may be irrelevant in one use and critical in another.
Uncertainty	Every timing claim needs clock source, uncertainty, and common-mode dependency metadata.
Statistics	Replace universal replication counts with test-specific analysis plans.
Governance	Lock review membership, quorum, pause authority, and ONESTAMP trigger.
CIE handling	Adopt unique-ID, preserve, no same-run promotion, and post-run disposition.
Human subjects	Keep MVP telemetry-only unless formal determination authorizes participant research.
Privacy	Packet capture only on owned/authorized systems and isolated test networks.
Software QA	Version-controlled repository, seed policy, configuration hash, defect log, and replayable artifacts.
Canon hygiene	Label established science, internal OAL architecture, hypothesis, and speculative extension separately.
Artifact authority	Choose one authoritative home for executable scenario cards and retire superseded copies.

19. Alternative Analysis and Red-Team Findings

19.1 "This is just signal processing"

Strict aliasing is mature signal processing; the project should not claim novelty for Nyquist. The research contribution is the cross-layer question: how sampling error, clock semantics, transport, presentation smoothing, human temporal integration, and root inaccessibility interact in one reconciliation stack.

19.2 "This is only perception science"

A perception-only program omits device clocks, buffering, causal history, network synchronization, camera acquisition, and correction. A systems-only program omits the final observer whose action depends on the presentation. The research object is the interaction between them.

19.3 "The brain takes discrete snapshots"

This remains a hypothesis family, not an architectural fact. Attention-dependent sampling research is important but contested, and the 2026 delay-field model offers a different mechanism. [E6][E7]

19.4 "If two clocks agree, the timeline is correct"

Agreement without independence can be a common-mode failure. Two logs synchronized to the same wrong reference can agree perfectly. Clock-source and failure-domain independence must therefore be part of every material claim.

19.5 "The canonical ledger solves ground truth"

Only for experiments where the ledger is outside the mechanism under test. If the campaign investigates root-ledger failure, the ledger cannot validate itself. This is exactly where an external oracle is required.

19.6 "Any mismatch proves deception"

No. PSIF correctly treats deception as one possible explanation after mechanism and evidence are established. Timing mismatch can arise from instrumentation, stale state, ordinary buffering, structural design, drift, or emergent interactions. [I3]

19.7 "Open Window must be literally true for the project to matter"

No. Even if every speculative cross-environment hypothesis is null, the temporal program remains useful for simulation fidelity, multi-device UX, incident reconstruction, distributed systems, cameras/displays, robotics, safety interfaces, and human factors. That independence strengthens the program.

19.8 "More days and more hardware mean more truth"

Not necessarily. Scope inflation can increase uncontrolled variables faster than evidence quality. A four-card lean program with independent timing and preregistered falsification is more informative than a 14-day maximal campaign whose nulls, clock independence, and provenance are unresolved.

20. What Works, What Does Not, and What Remains Unresolved

20.1 What works

- The user-originated stack intuition: mind + devices + systems each reconcile partial temporal streams.

- Open Window v0.3 as a disciplined baseline: declared interfaces, optional airlock, benign proxies, CETE above infection, candidate-event restraint.
- PSIF evidence discipline and the separation of presentation, state, mechanism, causation, and intent.
- CFTRL as the inaccessible-root stress question and Department of Endings as the run/log/learn operational function.
- The existing temporal draft's taxonomy, L0-L5 model, metric dictionary, gates, Python slice, and lean-first funding logic.
- TA-01 as the first true alias benchmark because it can make a numerical prediction and can be falsified by a prefilter/sample-rate intervention.

20.2 What does not work

- Using temporal aliasing as the name for every timing discrepancy.
- Treating the brain as a universal fixed-rate sampler.
- Calling clock disagreement a CETE by default.
- Treating a complete-looking ledger as infallible during root-failure experiments.
- Using a time-normalization layer that overwrites raw timestamps.
- Assuming multiple devices are independent merely because they are separate physical devices.
- Using human-participant trials before the telemetry and timing harness is stable.
- Promoting a vivid one-off demonstration into doctrine or canon.
- Letting multiple competing "final" documents remain active without an authority hierarchy.

20.3 Still unresolved

- Whether Temporal Integrity becomes a formal umbrella document or remains an annex to the current temporal aliasing draft.
- Whether the L0-L5 timing diagram is accepted only for timing work or promoted to a broader Open Window architecture.
- Exact review-group membership, quorum, and pause authority.
- Authoritative repository/annex for TA/TI scenario cards.
- The T1 standard for clock independence and uncertainty.
- Whether the Temporal Endgame Desk is Department of Endings canon or a lab-only function.
- Whether human reports are deferred entirely to a separate research protocol.
- Whether CFTRL includes canonical-ledger corruption as a formal root-failure class.

21. Questions for Architect Review

1. Do you want Temporal Integrity / Temporal Presentation-State Divergence to become the umbrella construct, with temporal aliasing below it?
2. Is "first-person operational truth" an acceptable replacement for stronger "objective truth" language in research-facing documents?
3. Should L0 be physically/logically outside the simulation under test whenever CFTRL is evaluated?
4. Should non-alias cards be renamed from TA to TI so the scenario ID itself preserves the taxonomy?
5. Should the Temporal Endgame Desk be Department of Endings canon or remain a laboratory implementation?
6. Should T1 remain telemetry-only, with all human perception work placed in a separately governed campaign?
7. What artifact is authoritative for executable cards: v0.3 annex, scenario packet, or controlled repository?

8. What level of clock-source independence is mandatory at T1, and what limitations are acceptable until T2?
9. How should materiality be defined by use case: absolute milliseconds, order inversion, missed event, or decision consequence?
10. May a time-normalization airlock buffer and translate timestamps if raw originals are immutable and preserved?
11. Should root failure explicitly include failure/corruption of the canonical event ledger itself?
12. Should Coherence-Correctness Separation become a headline CFTRL metric?
13. Should the current Grok temporal draft be revised to v0.2, or should this independent pass become a separate comparison document?
14. Which internal source is the governing baseline when historical layer diagrams conflict?
15. What external-use narrative is prioritized first: scientific instrumentation, simulation QA, HCI/human factors, or incident reconstruction?

22. Immediate Action Plan

Phase	Action	Exit evidence
A - Architecture lock	Approve terminology: Temporal Integrity/TPSD, strict TA definition, L0-L5 timing stack, raw-vs-derived timestamp rule.	Signed review disposition; no conflicting active definitions.
B - T0 software	Create versioned Python timing repository and minimum objects; implement fold, skew, buffer, wall-step, root-maintenance faults.	Reproducible seeded runs; JSON provenance records; defect log.
C - Calibration	Execute TA-00 and characterize timestamp uncertainty/common-mode dependencies.	G1 instrumentation pass or explicit limitation.
D - Strict alias benchmark	Execute TA-01/TA-02 with preregistered mathematical prediction and intervention.	Predicted fold reproduced or clean falsification.
E - Neighbor mechanisms	Execute TI-03 through TI-06 and classify skew, buffer, rolling-shutter, and clock-semantic effects separately.	Mechanism-specific metrics; replay.
F - Real-world bench	Run RW-01 through RW-04 with independent channels and preserved raw traces.	Bench recurrence; uncertainty report.
G - CFTRL timing	Run root-maintenance loss and common-mode oracle failure tests.	Scale-to-failure curve; CCS; redundancy comparison.
H - Correction	Apply causal IDs, monotonic time, anti-alias filters, age-of-data, and normalization controls.	G7 correction persistence evidence.
I - Funding decision	Decide whether results justify T2 instrumentation and/or Project Pitch.	Go / no-go funding memo.
J - Human campaign	Only after formal human-research determination and stable instrumentation.	Approved protocol; participant-level analysis plan.

22.1 Near-term priority

Recommended next artifact

Do not create another broad "unified doctrine" rewrite. Produce a focused timing addendum or v0.2 correction pass that adopts the strict-aliasing distinction, Temporal Provenance Overlay, TA-00 calibration,

independent-oracle rule, revised scenario IDs, and the implementation blockers in this report.

23. Final Synthesis

The strongest concept emerging from this work is not that every brain or device is wrong. It is that every observer and every technology layer offers a representation whose temporal fidelity must be evaluated relative to the decision being made. Cameras reconcile light into exposures and frames. Networks reconcile asynchronous clocks. Operating systems maintain multiple time bases. Media systems buffer and refresh. User interfaces decide what to label as current. Human perception binds, postdicts, attends, and preserves continuity. Organizations then reconcile all of those records into incident histories and narratives.

This creates a fundamental systems question: when does the unified view remain useful but become materially different from the events that generated it? A system may display every event and still misrepresent order. It may contain accurate timestamps and still have the wrong clock source. It may show a smooth video whose motion is aliased. It may show a "live" state that is buffered. A person may experience simultaneity where the evidence records asynchrony. None of those outcomes alone proves deception; all of them are testable presentation-state questions.

Temporal Integrity therefore gives the existing OAL architecture a practical research spine. Strict aliasing supplies a mathematically predictable failure. PSIF supplies evidence discipline. Open Window supplies interface conditions and candidate-event restraint. CFTRL supplies the inaccessible-root question. The Department of Endings supplies the run-log-learn operational posture. The resulting program can be useful even if every speculative hypothesis fails because the laboratory is fundamentally testing synchronization, sampling, provenance, causal reconstruction, human-machine presentation, and correction.

Compact rule

Do not mistake a coherent timeline for a verified timeline. Preserve raw time. Identify the clock. Quantify uncertainty. Reconstruct causal order. Predict strict aliases before observing them. Intervene on the mechanism. Replay. Recur. Correct. Re-test.

Appendix A. Mandatory Scenario-Card Schema

- Scenario ID and revision
- Scenario title
- Purpose / decision supported
- Scientific classification (TA/SK/BF/etc.)
- Declared Interface Object(s)
- OAL proxy class(es), if applicable
- Airlock condition: absent / present / varied
- Exact setup and initial-state requirements
- Intended use and claim ceiling
- Null hypothesis / falsification condition
- Positive control
- Negative control
- Independent timing reference and fault domain
- Clock source(s) and uncertainty
- Primary metrics
- Secondary observations
- Materiality threshold
- Stop conditions
- Required replications / analysis plan
- Replay requirement
- Independent recurrence requirement
- Intervention / severance test
- Correction test
- Data products to preserve
- Raw/derived timestamp handling
- Privacy / human-subjects constraints
- Candidate Interface Event rule
- Review-group disposition authority
- Post-run classification: supported / inconclusive / falsified / blocked / deferred
- Known limitations and unresolved questions

Appendix B. Minimum Temporal Provenance Record

Example logical record for every material event:

```

event_id
parent_event_id
condition_id
source_id
process_id
source_time_raw
capture_start_time
capture_end_time
local_monotonic_time
local_wall_time
ingest_time
decode_time

```

queue_release_time
 presentation_time
 participant_report_time (if applicable)
 clock_id
 clock_type
 clock_reference
 clock_uncertainty
 last_sync_state
 sequence_number
 dependency_ids
 raw_hash
 software_version
 configuration_hash
 random_seed (if applicable)
 normalized_time
 normalization_method
 derived_timestamp_flag
 evidence_label

Appendix C. Research Vocabulary

Term	Working definition
Temporal Integrity	Degree to which presented timing, order, frequency, duration, and simultaneity correspond to best-supported event history for the intended use.
Temporal Presentation-State Divergence (TPSD)	Material inconsistency between presented timeline and evidenced timing/order/duration/causal history.
Temporal aliasing	Strict sampling-driven fold in which higher-frequency structure reconstructs as lower-frequency or reversed content.
Nyquist Margin (NM)	Laboratory headroom measure $f_s/(2*B)$ for defined band-limited content.
Reconciler	Any process that emits one usable timeline from multiple streams.
First-person operational truth	The participant-accessible representation from which the participant acts.
Simulation ground state	Authoritative event/state history for a defined simulation experiment, unless itself under test.
Audit reference / oracle	Independent evidence channel intentionally placed outside the relevant fault domain.
Time-normalization airlock	Optional interface control that validates, buffers, translates, or refuses timestamps under a declared skew policy while preserving raw originals.
Coherence-Correctness Separation	Condition in which presentation remains smooth/coherent while temporal fidelity degrades.
Root-inaccessible timing failure	CFTRL case in which lower layers cannot inspect or repair timing/sampling policy of an upstream/root layer.
Candidate Interface Event	Open Window v0.3 category for an unexplained possible undeclared relationship requiring preservation and later validation, not same-run promotion.

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Source-use note

External sources are used to establish or constrain scientific and technical claims. Internal OAL sources are used to preserve the project's terminology, locked decisions, incorporation goals, and governance history. The report intentionally does not use external literature to convert speculative OAL constructs into claims of demonstrated physical phenomena.